

Meeting-report

A Simple and Accurate Approach for Thickness Measurement of Particles and Thin Films Using SEM-EDS

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Characterisation of thin samples using scanning electron microscopes (SEM) equipped with an energy dispersive spectrometers (EDS) presents both a considerable interest and a challenging task. One of the major difficulties is mainly associated with the lack of knowledge of the sample size in the direction of the incident electron beam (Z direction) as imaging is only performed in the X–Y plane. The determination of the thickness of samples often turns out to be an important aspect during both imaging with SEM and chemical analysis with EDS [1]. Several accurate approaches for thickness measurement using X-ray microanalysis were developed [2–5]. As these approaches were only tested on samples with a thickness ranging from few to hundreds of nanometers, their adaptability to micro-sized samples is not guaranteed. Another inconvenience that is worth mentioning is that most of the available thin-film programs dedicated to thickness measurement are not free [6–9].

Thickness measurement can be carried out efficiently using transmission electron microscopy (TEM) based techniques, such as energy-loss spectroscopy (EELS) [10,11]. Although fast and accurate thickness calculation can be carried out with EELS, its accuracy depends on the knowledge of complex parameters such as the optical refractive index and the inelastic mean free path that are difficult to evaluate for many materials [12]. Moreover, multiple complex factors should be taken into account with regards to the application of TEM-based techniques (e.g. sample preparation and sensitivity of image analysis to the thickness of the studied specimen).

Unlike TEM, samples can be easily imaged in SEM. Methods allowing for fast and accurate thickness measurement using SEM can therefore be very optimal. In this work, we present a simple, fast and accurate approach for thickness measurements using the combined use of SEM and EDS. The method was previously validated on K411 glass micro-particles (Fig.1.b) as well as particles composed of U-Ce oxide exhibiting a well-defined geometry and a size ranging from 200 nm to 4000 nm [13]. In order to assess the validity of our approach and widen its applicability on a different set of materials, thickness measurement was carried out on thin films with sizes ranging from 100 to 800 nm. The results were afterwards compared to measurements performed with interferometric microscopy (IM).

The presented approach is applicable on samples with a thickness smaller than the electron depth range and does not require any knowledge of instrumental parameters. The thicknesses were determined from the X-ray intensity associated with the $K\alpha$ line of the substrate on which thin films were deposited. This was achieved by probing each sample using a focused electron beam with an energy high enough for the transmission of electrons to occur. After being transmitted through the thin sample, electrons strike the substrate at a sufficient energy and induce the emission of characteristic X rays (Fig. 1.a). The X-ray intensity emitted from the substrate is proportional to the energy of transmitted electrons and was used to determine the energy loss of electrons. This enabled us to calculate the distance that electrons travel inside the sample and therefore determine a local apparent thickness.

Thickness measurement of thin films composed of alanine deposited on a MgF_2 substrate was carried out using both IM (Fig 2) and SEM-EDS. The obtained results are very promising, showing an excellent agreement between the two techniques.

The approach presented in this work allows for thickness measurement on a wide thickness range and can provide useful information about local three-dimensional (3D) geometry of thin films and microparticles. In EDS, a prior knowledge of the thickness could provide insight into the amount of X-rays absorbed within the sample, which constitutes an important parameter for quantitative analysis as well as the interpretation of EDS maps. The presented method can also be very advantageous to methods dedicated to the measurement of optical properties as such properties are strongly dependent of the sample's morphology [14].

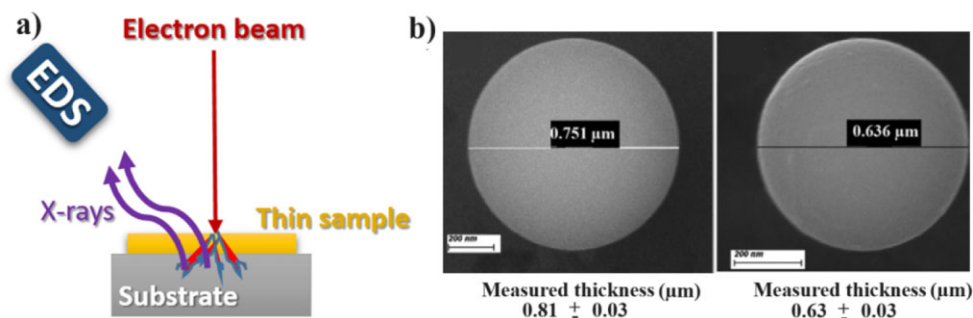


Fig. 1. a) X-ray emission of the substrate induced by electron transmission through a thin sample. b) two K411 glass micro-particles for which the thickness was measured using the SEM-EDS method [13].

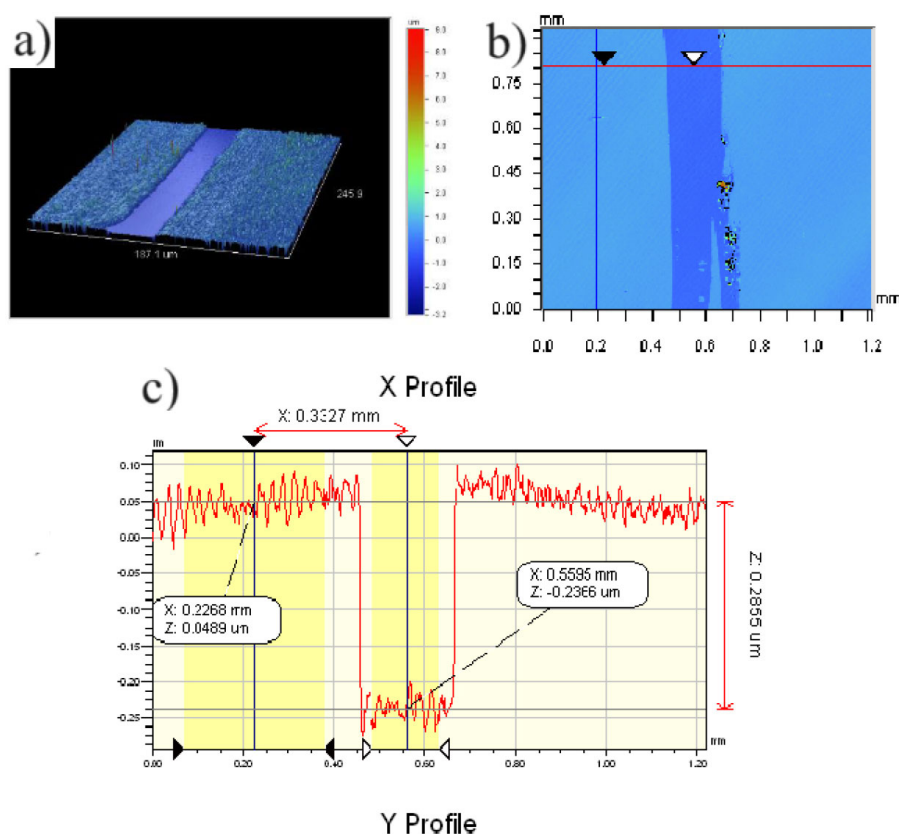


Fig. 2. Thickness measurement using interferometric microscopy: a), b) 3D and 2D views of the thin film deposited on a MgF_2 substrate, respectively, with an induced discontinuity at the middle of the deposition. c) Profiles in the X direction of the thin film and the discontinuity allowing for thickness measurement [14]. The discontinuity was induced manually after the deposition by removing the deposition at the middle using a needle.

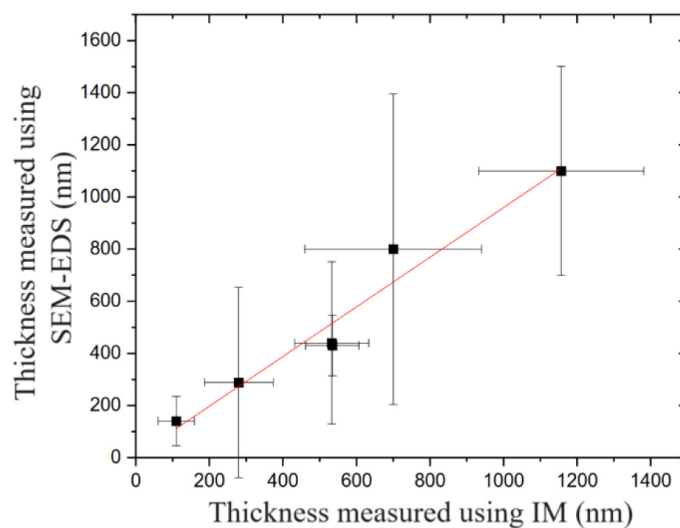


Fig. 3. Comparison between thickness measurements using SEM-EDS and IM [14].

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